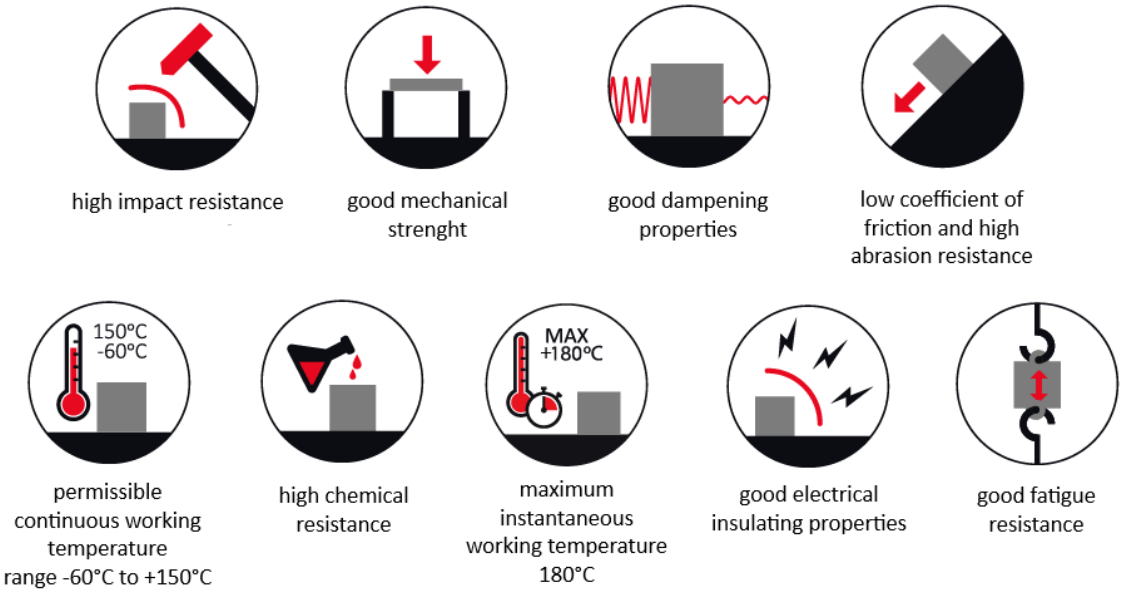
NOCTUO

PRODUCT CARD

NYLON PA6



1. GENERAL INFORMATION ABOUT THE PRODUCT

Nylon PA6 (Poliamid) is a high-quality, general-purpose, thermoplastic construction material. It has good mechanical and chemical parameters. Nylon has high stretching, bending and abrasion resistance. It has high hygroscopicity, which requires very careful storage and might causes necesity of drying the material before printing. This attribute allows easy dyeing after printing.

Main Nylon features:

- very high mechanical strength,
- high impact resistance,
- high ability to damp mechanical vibrations,
- good fatigue strength,
- very good sliding properties, abrasion and scratch resistance and low coefficient of friction,
- high warm resistance, permissible continuous operation temperature between -60°C and +150°C,
- high chemical resistance, especially to organic solvent, including oils, greases and fuels,
- good electrical insulation properties.

2. TECHNICAL PARAMETERS

CHARACTERISTICS	TEST METHOD	TEST CONDITIONS	IU	VALUE	
	ISO			SOLID STATE	CONDITIONED STATE
PHYSICAL					
Melt temperature	11357-1-3	10°C/min.	°C	221	
Density	1183	-	g/cm^3	1,14	
Moisture absorption	62	23°C/50%RH	%	3	
Water absorption to saturation	62	23°C/sat.	%	9,5	
Processing shrinkage II/⊥	294-4	60x60x2	%	1,4/1,4	
MECHANICAL					
Yield strength	527-1,-2	50mm/min	MPa	78	45
Elongation at break	527-1,-2	50mm/min	%	>100	>250
Elastic modulus at tension	527-1,-2	1mm/min	MPa	2800	1100
Bending stress	178	2mm/min	MPa	82	28
Elastic modulus at bending	178	2mm/min	MPa	2400	900
Notched Charpy impact strength	179-1	1eA	kJ/m^2	6	13
Notched Charpy Impact strength (-30°C)	179-1	1eA	kJ/m^2	3	3
THERMAL					
Vicat softening point	306	50N	°C	195	180
Deflection temperature under load	75-1,-2	1,8 MPa	°C	60	50
Coefficient of linear thermal expansion II/⊥	11359-1/-2	23°C - 55°C	E-4/°C	1,1/1,2	
Temperature index	IEC 60216	20000 h 5000 h	°C °C	70 85	
Maximum instantaneous working temperature	-	-	°C	<180	
COMBUSTIBILITY					
Flammability level test	UL94	3,2 mm	Class	HB	
Flammability index of incandescent material wire(GWFI)	IEC-60695-2-12	2 mm	°C	750	
ELECTRICAL					
Surface resistivity	IEC 60093	-	Ω	10^{15}	10^{13}
Volume resistivity	IEC 60093	-	Ωxcm	10^{15}	10^{12}
Dielectric strength	IEC 60243-1	2mm	kV/mm	21	24
Dielectric constant	IEC 60250	1MHz	-	3,5	4,2
Dielectric loss factor	IEC 60250	1MHz	E-4	220	2000
Comparative Tracking Index	IEC 60112	solution A	V	600	600

Dry condition - moisture content max. 0.2%
Conditioned condition - fittings conditioned at 23°C and 50% relative humidity until equilibrium
Tests have been done in 23°C, if it's not marked differently.

3. RECOMMENDATION OF PRINTING

The hygroscopicity of nylon must be considered. Most problems with printing is caused by moisture filament. You may find some issues with proper adhesion too. Using perforated bed, PVA glue, specialized product or coating is recommended.
Nylon has relative high processing shrinkage. It could be prevented, depending on the printer capabilties, the parameters of the printed model or expectations, by manipulating chamber and bed temperature and increasing the amount of served filament.

Recommended parameters of printing:

Hotend temperature	265 - 290 °C
Bed temperature	> 100 °C
Print speed	< 100 mm/s

Drying:

Vacuum or freeze drying if possible. In home conditions, heat soaking in temperature of about 80 - 90°C with thermocirculation for 4-6 hours. The highest drying efficiency is achieved by unwinding the required amount of filament from the spool and drying loose coils.

Problem	Possible cause	Proposed solution
Foaming of the material during printing	- Filament moisture	-Drying
Sounds of 'shooting' of the steam during printing		
Uncontrolled curling of extruded part		
Weak layers adhesion	- Filament moisture - Extruder temperature is too low - Too high speed of printing	- Drying - Raising the extruder temperature - Reducing the print speed
Uneven feeding – losing steps of the feeder / filament sliding on the drive gear	- Extruder temperature is too low -Weak feader pressure	- Raising the extruder temperature - Raising the pressure
Print is peeling of the bed	- Improperly prepared surface - Processing shrinkage	-Raising the bed temperature -Degreasing the bed / using another source of adhesion / using perforated bed - Compensation with amount of served filament
Edge hemming	- Processing shrinkage	- Compensation with amount of served filament / selection cooling parameters / changing the chamber temperature
Color change - yellowing of the material	- Extruder temperature is too high	- Reducing the extruder temperature

4. SAFETY NOTES

Exhaust fan is recommended.
Air filters in printer is recommended.
Nylon needs to be used only in well ventilated conditions.
Inhaling fumes generated during the printing must be avoided.

Generating fumes during the printing depends mainly on printing temperature. In case of visibly raising emission level, the printing needs to end. Check the hotend temperature and efficiency of the control system before using it next time.

In proper using conditions, the product doesn't endanger health.

It’s forbidden to set fire or exceed decomposition temperature!

Decomposition of nylon is typically over 350 °C.
Main ingredient of decomposition is ε-caprolactam.

Detailed safety information available in SDS.